
Knowledge On-Off Switch: Flexible Unlearning and Relearning with O^3 Framework

Korea University COSE461 Final Project

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Abstract

Large Language Models (LLMs) increasingly require machine unlearning to comply with privacy regulations and copyright constraints. However, existing unlearning methods are effectively irreversible: restoring forgotten knowledge typically requires access to the retained dataset, which is frequently prohibited. We term this challenge *Machine Relearning* and define *Selectivity* as the ability to restore specific domains without affecting others, without relying on the retained data.

We propose a training-free relearning switch built on the O^3 continual unlearning framework. Rather than modifying model parameters or retraining the unlearning LoRA, our method introduces a score-space boundary shift into the inference-time OOD scoring pipeline. Non-restored domains are unaffected, and the switch is instantly reversible by removing the margin.

We evaluate on ScienceQA across four scientific domains. The proposed method achieves an ON Sample-level Recovery Rate (S.R.R.) of 97% and an ON Distribution-level Recovery Rate (D.R.R.) of 84.5%, demonstrating effective restoration without any fine-tuning. OFF domain recovery rates remain at -0.38% and -0.60% , confirming that non-restored domains stay unlearned. Recovery rates are invariant across all values of k —the number of simultaneously restored domains—providing evidence of domain disentanglement. General model utility, measured on three held-out benchmarks, is fully preserved across all restore configurations.

1 Introduction

Driven by the exponential growth of data and the requisite computational resources, Large Language Models (LLMs) have achieved unprecedented advancements in recent years. However, alongside this rapid progress, novel internal and external challenges have emerged, most notably hallucinations and jailbreaking [1, 2].

Now machine unlearning has also emerged as a crucial topic. Certain data must be removed—whether to prevent privacy infringements or protect systems from contamination from a security standpoint

[3]. Therefore, research has been initiated on techniques to eradicate the influence of unnecessary or harmful data from models, a process referred to as *Machine Unlearning*, or simply *unlearning*.

However, implementing machine unlearning in practice remains highly challenging. This difficulty arises because the learning and training processes are inherently stochastic and nondeterministic, making it unclear which specific data points are extracted to adjust model parameters. Furthermore, since model updates accumulate sequentially during training, it is impossible to simply apply an inverse operation to reverse the process [4].

To overcome these limitations, machine unlearning approaches can be broadly categorized along two major axes. The first axis concerns precision: Exact Unlearning, which algorithmically eradicates data influence entirely, and Approximate Unlearning, which minimizes influence through constrained updates [5]. The second axis pertains to the mechanism: Parameter Optimization, which modifies model weights, and In-Context Learning, which adjusts prompts or context to withhold specific information. Representative examples for each of the four resulting paradigms include S³T [6] for exact-parameter, ERASE [7] for exact-in-context, SalUn [8] for approximate-parameter, and Pawelczyk et al.’s methodology [9] for approximate-in-context unlearning.

Restoring previously forgotten knowledge is also required in practice—for instance, after resolving a copyright dispute or when an AI agent must dynamically toggle domain knowledge during deployment. We term this paradigm *Machine Relearning*, or *relearning* for short. However, attempting restoration in LLMs that undergo continuous unlearning realistically requires access to a fraction of the retained dataset [10]—a practice frequently prohibited due to privacy or copyright constraints. Therefore, it is imperative to selectively flash specific knowledge on the fly without relying on the retained dataset, a property we define as *Selectivity*.

Accordingly, this work aims to develop an inference-stage LLM capable of selectively toggling specific knowledge on and off instantaneously. The main contributions of this work include:

- We investigate a novel methodology to unlearn and relearn domain-specific knowledge with the O^3 framework, which leverages Orthogonal Low-Rank Adapters (LoRA) and an Out-Of-Distribution (OOD) detector.
- We propose a training-free relearning switch that restores selected domains at inference time by shifting the OOD boundary distance, with no parameter updates or access to the original data required, thereby achieving domain-level selectivity.

2 Related Work

2.1 OOD Detection

OOD detection has evolved from classical methods such as one-class SVM and density estimation to deep learning-based approaches [11], which identify inputs that deviate from the training distribution. Since the test samples originate from a distribution distinct from the training distribution, the model identifies this discrepancy and subsequently classifies the testing data as out-of-distribution (OOD). This capability has been incorporated into unlearning frameworks to selectively modulate model behavior at inference time.

2.2 O^3 Framework

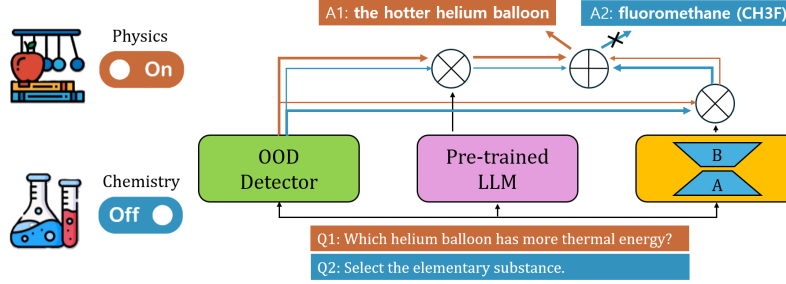
Most relevant to our work is the O^3 framework [12], which performs continual machine unlearning by combining orthogonal LoRA [13]-based parameter updates with an OOD detector. O^3 enforces orthogonality across successive unlearning requests to prevent interference between them, while the OOD detector estimates at inference time whether an input is related to a previously unlearned domain and modulates how strongly the unlearning adapter is applied. We use this as our baseline model and describe its inference mechanism in detail in Section 3.

3 Approach

Building on the O^3 continual unlearning framework, we propose a training-free relearning switch that modifies only the inference-time OOD scoring process. The goal is to restore selected previously

unlearned domains without updating the base model, retraining the unlearning LoRA, or storing the original forgotten data.

Figure 1: An overview of the knowledge on-off switch with the O^3 framework



Since our method operates inside the O^3 inference pipeline, we first clarify the implementation-level quantities used throughout this section. The OOD detector is a module that estimates whether an input resembles a previously unlearned domain. O^3 represents each input with layer-wise feature scores; these scores include a Mahalanobis-style distance, which measures how far the input features are from the feature statistics of the unlearning data.

A One-Class SVM (OCSVM) is then used to model the score region associated with the unlearning data. Its output is converted into an OOD weight, a scalar coefficient that determines how strongly the unlearning LoRA is applied during generation.

3.1 O^3 Soft-Weighted Inference

O^3 performs continual unlearning by attaching an orthogonal unlearning LoRA to a base LLM. The orthogonal regularization separates the parameter subspaces used by different unlearning requests, while task-specific OOD detectors estimate whether an input query is associated with previously unlearned knowledge. At inference time, O^3 computes a candidate OOD weight for each previous unlearning request and uses the maximum weight to decide how strongly the shared unlearning LoRA should be applied. In simplified form, the hidden representation is updated as follows:

$$h' = Wh + w(x)ABh, \quad (1)$$

where W denotes the original model weight, AB denotes the shared orthogonal unlearning LoRA update, and $w(x) = \max_t w_t(x)$ is the maximum OOD-based soft weight across previous unlearning requests. A larger $w(x)$ strengthens the unlearning effect, whereas a smaller $w(x)$ makes inference closer to the base model behavior.

In O^3 , each unlearning request can be viewed as a domain represented in the detector score space. The task boundary refers to the hypersphere learned by the OCSVM around the score region of the corresponding unlearning data. O^3 uses the distance between an input score and this boundary to estimate how strongly the input is associated with the unlearning target. The original O^3 paper describes this hypersphere boundary distance as

$$d_H^t(x) = |s(x)^t - c^t| - R^t, \quad (2)$$

where $s(x)^t$ is the score of input x for task t , c^t is the center, and R^t is the radius of the task boundary. This formulation suggests that changing the effective boundary could control whether the unlearning LoRA is activated.

3.2 From Radius Control to Boundary Shift

In the released O^3 implementation,¹ however, the radius R^t is not separately manipulated at inference time. The available inference signal can instead be summarized as a sequence from layer-wise feature scores to an OCSVM detector score and then to a candidate OOD weight:

¹<https://github.com/GCYZSL/O3-LLM-UNLEARNING>

$$\text{layer-wise feature score} \rightarrow z_t(x) \rightarrow w_t(x). \quad (3)$$

Intuitively, our goal is to make inputs from a restored domain appear less similar to the unlearning region in the OOD score space. If the OOD weight for that domain decreases, the unlearning LoRA is applied less strongly, allowing the base model’s original knowledge to influence inference again.

Therefore, rather than modifying R^t directly, we implement a score-space boundary shift using quantities already present in the O^3 scoring pipeline. Let $z_t(x)$ be the OCSVM score for task t , and let $b_t = x_0^t$ be the reference score used by the O^3 weight-mapping step. Intuitively, x_0^t plays the role of a center point in the one-dimensional OCSVM score space when detector scores are converted into OOD weights. We define the score-space distance as

$$d_t^{\text{score}}(x) = |z_t(x) - b_t|. \quad (4)$$

For a restore set $\mathcal{R}$, we add a restore margin only to the selected domains:

$$\Delta_t = \begin{cases} \lambda |x_0^t - \tau_t|, & t \in \mathcal{R}, \\ 0, & t \notin \mathcal{R}, \end{cases} \quad (5)$$

Here, x_0^t is the reference point for computing the OOD weight, and τ_t is the threshold that separates the unlearning score region from the non-target region. The distance $|x_0^t - \tau_t|$ can be viewed as a natural unit length in the score space of task t . Since OCSVM score scales differ across domains, we normalize the restore margin by this task-specific distance rather than adding the same raw margin to all domains. The hyperparameter λ controls how many such units are applied as the restore margin.

The shifted distance is

$$\tilde{d}_t^{\text{score}}(x) = d_t^{\text{score}}(x) + \Delta_t. \quad (6)$$

We then convert this shifted distance back to the OCSVM score space:

$$\tilde{z}_t(x) = b_t + \text{sign}(z_t(x) - b_t) (|z_t(x) - b_t| + \Delta_t). \quad (7)$$

Finally, the shifted score $\tilde{z}_t(x)$ is passed to the original O^3 weight-mapping step to compute the OOD weight for task t . This keeps the OOD detector and weighting pipeline intact while shifting selected domains away from the unlearning boundary in score space. Consequently, their OOD weights decrease, the unlearning LoRA is applied less strongly, and the base model’s knowledge becomes more accessible. For non-restored domains, $\Delta_t = 0$, so their original unlearning behavior is preserved.

3.3 Inference Procedure

For every input and every unlearned domain, the detector score is computed as in O^3 ; selected restore domains simply receive an additional boundary shift—controlled through $\mathcal{R}$ and λ —before the OOD weight is computed.

For each input x , inference proceeds as follows:

1. Compute the Mahalanobis feature score and the OCSVM score $z_t(x)$ for each unlearned task t .
2. If $t \in \mathcal{R}$, apply the restore margin Δ_t and obtain the shifted score $\tilde{z}_t(x)$. If $t \notin \mathcal{R}$, keep the original score unchanged.
3. Feed the resulting score into the original O^3 weight-mapping step to obtain the task-specific candidate weight $w_t(x)$.
4. Select the maximum candidate weight $w(x) = \max_t w_t(x)$ and use it to control the strength of the shared unlearning LoRA during generation.

Thus, relearning is implemented by modulating the OOD weight at inference time rather than by adding a new training stage. The same checkpoints can support different knowledge states simply by changing the restore set.

4 Experiments

4.1 Data

We evaluate the proposed method under the ScienceQA benchmark [14] using the O^3 repository setting. The main unlearning and relearning evaluation uses four ScienceQA domains: biology, physics, chemistry, and economics. Following the original O^3 ScienceQA setup, the continual unlearning sequence is biology $\rightarrow$ physics $\rightarrow$ chemistry $\rightarrow$ economics. Each example is treated as a multiple-choice question, and the model output is compared with the correct answer option. To measure utility preservation, we additionally evaluate on retained ScienceQA domains, CommonsenseQA, and OpenBookQA. The retained ScienceQA split tests whether relearning damages nearby in-domain knowledge, while CommonsenseQA and OpenBookQA test whether general QA ability is preserved.

4.2 Evaluation Method

We evaluate both relearning effectiveness and utility preservation. For unlearning, we report accuracy on the unlearning train and test splits. Following the O^3 evaluation protocol, S.U. denotes sample-level unlearning on the train split, and D.U. denotes distribution-level unlearning on the test split. Lower S.U. and D.U. indicate stronger unlearning.

For relearning, we distinguish between ON and OFF domains. ON domains are included in the restore set $\mathcal{R}$, while OFF domains are not restored and should remain unlearned. We report sample-level relearning on train data and distribution-level relearning on test data. To compare domains with different base accuracies and different unlearned accuracies, we use the recovery rate

$$R.R. = \frac{acc_{current} - acc_{unlearned}}{acc_{base} - acc_{unlearned}}. \quad (8)$$

For ON domains, a recovery rate closer to 1 (or 100% when reported as a percentage) indicates stronger restoration. For OFF domains, a recovery rate close to zero is preferred, because it indicates that non-restored domains remain unlearned. We also report utility on R.D. (retained ScienceQA domains), C.QA. (CommonsenseQA), and O.QA. (OpenBookQA). Unlike the recovery metrics, R.D., C.QA., and O.QA. are reported as raw accuracies. Here acc_{base} is the accuracy before unlearning, $acc_{unlearned}$ is the accuracy of the fully unlearned $k = 0$ model, and $acc_{current}$ is the accuracy under a given restore setting.

4.3 Experimental Details

We use the released O^3 ScienceQA checkpoint, gcyzsl/03_LLAMA2_ScienceQA,² as the base model checkpoint. We keep the same base model, unlearning LoRA checkpoints, and OOD detector checkpoints as in the O^3 ScienceQA pipeline; only the inference-time OOD scoring logic in ScienceQA_expriment/eval_o3.py is modified. All reported boundary-shift relearning results use $\lambda = 2$, corresponding to restore_delta_scale=2. All experiments are run with seed 0.

The unlearning checkpoint used in our experiments was prepared with a reduced training configuration: num_epochs=1 and batch_size=16. This setting is lighter than the full O^3 training configuration and is discussed further in the analysis section. Importantly, the proposed relearning switch itself does not introduce any additional training stage.

The baseline model and the $k = 0$ fully unlearned model are evaluated separately and used as reference points for the recovery-rate computation. The boundary-shift evaluations cover all restore combinations for $k = 1$ through $k = 4$, yielding 15 relearning configurations in total. We report the average recovery rate across cases with the same k .

²https://huggingface.co/gcyzsl/03_LLAMA2_ScienceQA

4.4 Results

Table 1 reports the per-domain $k = 0$ unlearned accuracies used as reference points for recovery-rate computation. Before applying the relearning switch, the four-domain unlearned model achieves an average S.U. of 7.61% and D.U. of 28.14%, showing that the targeted domains are substantially unlearned. Table 2 summarizes the relearning and utility results. All S.R.R. and D.R.R. entries are recovery rates reported as percentages, while R.D., C.QA., and O.QA. are accuracy values.

Table 1: Per-domain $k = 0$ fully unlearned accuracies used for recovery-rate computation. The average column reports the mean across four domains.

	Biology	Physics	Chemistry	Economics	Average
$k = 0$ train	6.96	4.03	7.20	12.24	7.61
$k = 0$ test	23.93	25.52	28.57	34.52	28.14

Table 2: Boundary-shift relearning results with $\lambda = 2$ (restore_delta_scale=2), averaged over all restore combinations with the same k . S.R.R. and D.R.R. denote sample-level and distribution-level recovery rates, respectively. R.D., C.QA., and O.QA. denote retained ScienceQA, CommonsenseQA, and OpenBookQA accuracy.

k	Cases	Restore setting	ON S.R.R. (%) $\uparrow$	ON D.R.R. (%) $\uparrow$	OFF S.R.R. (%) $\downarrow$	OFF D.R.R. (%) $\downarrow$	R.D.	C.QA.	O.QA.
0	1	all domains unlearned	—	—	0.00	0.00	91.34	79.20	83.20
1	4	single-domain restore avg.	97.00	84.50	-0.38	-0.60	91.34	79.20	83.20
2	6	two-domain restore avg.	97.00	84.50	-0.38	-0.60	91.34	79.20	83.20
3	4	three-domain restore avg.	97.00	84.50	-0.38	-0.60	91.34	79.20	83.20
4	1	all domains restored	97.00	84.50	—	—	91.34	79.20	83.20

Table 2 reports the quantitative results. Across restore settings, ON recovery rates are high, while OFF recovery rates remain close to zero. The utility metrics R.D., C.QA., and O.QA. remain unchanged across all restore settings. We provide a detailed interpretation of these trends in the analysis section.

5 Analysis

5.1 Unlearning Performance

Prior to evaluating relearning performance, we verify that the base unlearning stage effectively suppresses domain-specific knowledge across all four target domains. Table 3 summarizes the train- and test-set accuracy of each method after sequentially unlearning biology $\rightarrow$ physics $\rightarrow$ chemistry $\rightarrow$ economics.

Table 3: Unlearning performance comparison. For the baseline (pre-unlearning), values represent domain accuracy before any unlearning is applied. For unlearned models, Train Acc. and Test Acc. correspond to S.U. and D.U., respectively.

Method	Train Acc. (%) $\downarrow$	Test Acc. (%) $\downarrow$
Baseline (pre-unlearning)	99.75	98.23
O ³ [12]	9.3	14.0
Ours	7.61	28.14

The baseline model achieves near-perfect accuracy on the four target domains prior to unlearning, with an average train-set accuracy of 99.75% and test-set accuracy of 98.23%. After unlearning, S.U. drops to **7.61%** and D.U. drops to **28.14%**, representing reductions of 92.14 and 70.09 percentage points from baseline, respectively. These results confirm that the unlearning LoRA successfully suppresses domain-specific knowledge at both the sample and distribution levels.

Compared to the O³ baseline (S.U.: 9.3%, D.U.: 14.0%), our S.U. is lower while our D.U. is higher, attributable to the reduced training configuration (num_epochs=1, batch_size=16) adopted in our experiments. Despite this gap, the dramatic reductions from baseline confirm that unlearning is effective and provide a valid foundation for the relearning experiments.

5.2 Relearning Effectiveness

As shown in Table 2, the ON Sample-level Recovery Rate (ON S.R.R.) reaches **97%** across all configurations with $k \geq 1$, demonstrating that the proposed modification to the boundary distance formula effectively restores unlearned knowledge on training data. The ON Distribution-level Recovery Rate (ON D.R.R.) of **84.5%** on held-out test samples further confirms that the restoration generalizes to unseen instances, indicating that the relearning mechanism captures the underlying domain distribution rather than memorizing specific training examples.

The gap between ON S.R.R. (97%) and ON D.R.R. (84.5%) reflects a natural train-test generalization difference. Notably, this gap arises without any additional fine-tuning or retraining of the LLM; the relearning is achieved purely by adjusting the boundary distance computation at inference time, making the approach highly parameter-efficient.

5.3 Selectivity of Domain-Specific Relearning

Selectivity—the ability to restore a target domain without affecting others—is the central claim of this work. We verify selectivity along three axes: ON domain recovery, OFF domain isolation, and consistency across k values.

ON Domain Recovery. Across $k = 1, 2, 3$, and 4, the ON S.R.R. and ON D.R.R. consistently reach 97% and 84.5%, respectively. This confirms that the relearning mechanism successfully restores knowledge for each designated domain, regardless of which or how many domains are simultaneously relearned.

OFF Domain Isolation. For domains not designated for relearning, the OFF S.R.R. and OFF D.R.R. are -0.38% and -0.60% , respectively—negligibly close to zero across all configurations with $k = 1, 2$, and 3. The slight negativity falls within numerical noise and does not indicate any meaningful leakage of knowledge into unlearned domains. This demonstrates that activating the relearning signal for domain t has no detectable effect on the unlearning state of domains $t' \neq t$.

k-Independence. The ON and OFF metrics are identical across $k = 1, 2, 3$, and 4. Whether one or three domains are relearned simultaneously, each domain recovers at the same rate and the remaining unlearned domains stay at the same level of forgetting. This k -independence provides strong evidence of domain disentanglement: the relearning adjustment for each domain operates independently within the OOD boundary distance space, with no cross-domain interference.

5.4 Utility Preservation

We use the $k = 0$ configuration (all domains unlearned, none relearned) as the utility baseline. At $k = 0$, the model achieves R.D. = **91.34%**, C.QA. = **79.2%**, and O.QA. = **83.2%**. As shown in Table 2, these values remain exactly unchanged across $k = 1, 2, 3$, and 4. This perfect stability confirms that the relearning procedure introduces no side effects on general model capability.

The result is expected by design: since only the boundary distance formula (Eq. 2) is modified at inference time, the LLM weights and unlearning LoRA parameters are never altered during the relearning phase. The modification adjusts only the soft weight $w(x)$, which controls how much the unlearning LoRA is applied to each input. Knowledge that was never subject to unlearning is therefore entirely unaffected by the relearning procedure, regardless of how many domains are restored.

6 Conclusion

6.1 Summary of Major Findings

This work extends the O^3 continual unlearning framework [12] with a domain-selective relearning capability, achieved by modifying a single inference-time formula—the boundary distance function (Eq. 2)—while leaving all other components unchanged. Experiments on ScienceQA across four continually unlearned domains yield three principal findings: (1) the proposed switch effectively

restores unlearned knowledge (ON S.R.R. 97%, ON D.R.R. 84.5%) without any additional fine-tuning; (2) non-restored domains remain unlearned (OFF rates of -0.38% and -0.60%), with recovery rates invariant across all k values, establishing domain disentanglement; and (3) general model utility is fully preserved across all restore configurations.

6.2 Theoretical and Practical Implications

Theoretical Implications. The findings of this work carry several theoretical implications for the design of controllable unlearning frameworks. First, the soft-weighted inference mechanism, originally conceived to modulate unlearning strength in a single direction—from the pre-trained model toward the unlearned state—proves to be bidirectionally controllable by adjusting the boundary distance formula (Eq. 2).

Second, the fact that relearning is achieved through a modification to a single formula highlights the modularity of the O^3 architecture. New capabilities can be introduced without redesigning the broader pipeline, suggesting that the framework is extensible by targeted intervention at well-defined components.

Practical Implications. From a practical standpoint, this work demonstrates that machine unlearning need not be an irreversible operation. By retaining the unlearning LoRA and OOD detector after the unlearning phase, the system can restore domain-specific knowledge on demand without any retraining, offering a reversible alternative to conventional unlearning approaches that permanently alter model parameters.

The domain-level granularity of the relearning mechanism further enables fine-grained knowledge management. Individual domains can be independently activated or deactivated at inference time without maintaining multiple model copies. This capability may be particularly relevant in scenarios where knowledge requirements vary across users, regions, or time periods—for instance, in settings where data usage regulations differ by jurisdiction, or where a previously restricted domain becomes permissible following a change in policy.

Finally, since the mechanism operates entirely at inference time, it introduces no training overhead and is fully compatible with existing O^3 deployments.

6.3 Limitations and Future Work

Limitations. Despite the promising results, several limitations should be acknowledged. The most notable is the gap between our D.U. (28.14%) and that of the O^3 baseline (14.0%). This discrepancy stems from the reduced training configuration adopted in our experiments, which limits both the distribution-level generalization and the statistical reliability of the reported results. Consequently, the unlearning effectiveness demonstrated in this work is weaker than that of the original O^3 framework.

The scope of evaluation is also limited. Experiments are conducted solely on ScienceQA with four academic domains, and only a single model (LLaMA2-7B) is evaluated. It remains an open question whether the proposed relearning mechanism generalizes to other datasets, domain types, or model architectures. Furthermore, our experimental setup assumes a strictly sequential protocol in which all domains are first unlearned before any relearning is performed. The more realistic scenario, in which unlearning and relearning requests arrive in an interleaved or concurrent fashion, is not addressed.

Finally, the relearning capability is contingent on the preservation of the unlearning LoRA and the OOD detector after the unlearning phase. If either component is discarded or updated, relearning can no longer be performed. In deployment settings where storage is constrained or where the unlearning pipeline is designed to minimize retained artifacts, this dependency may pose a practical burden.

Future Work. Several directions present themselves for future investigation. The most immediate is to re-run the unlearning stage under a more complete training configuration—using more epochs, a larger batch size, and multiple seeds—to produce a stronger baseline and statistically reliable estimates. Beyond replication, systematic evaluation across broader datasets, domain types, and model architectures would establish the generalizability of the proposed mechanism; and extending the framework to handle interleaved unlearning and relearning requests would bring it closer to real-world deployment conditions.

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A Appendix: Team contributions

Taehyun Kim (2021320052): Conceptualization, literature review and baseline selection, evaluation criteria establishment (Recovery Rate, ON/OFF S.R.R./D.R.R. framework design), experimental environment setup, initial model implementation and evaluation, and report writing.

Taekyu Park (2021320080): Conceptualization, literature review and baseline selection, boundary-shift relearning switch implementation, hyperparameter tuning, full model evaluation (unlearning, relearning, and utility preservation), and report writing.

Yejun Oh (2021320312): Conceptualization, literature review and baseline selection, experimental environment setup, initial model implementation and evaluation, full model evaluation (unlearning, relearning, and utility preservation), and report writing.

B Appendix: Additional experimental results

Table 4: Baseline accuracy (R.D.) per category (n = sample size)

Category	Acc (%)	Category	Acc (%)
Capitalization	100.00 (33)	Pronouns	100.00 (2)
Civics	92.31 (26)	Punctuation	98.00 (100)
Culture	100.00 (2)	Reading Comprehension	87.50 (8)
Earth Science	98.46 (65)	Reference Skills	63.89 (144)
Figurative Language	97.00 (267)	Science & Engineering	100.00 (61)
Geography	100.00 (4)	Units and Measurement	95.17 (207)
Grammar	95.71 (70)	US History	66.67 (6)
Literacy in Science	100.00 (2)	Verbs	100.00 (52)
Phonological Awareness	35.29 (17)	Vocabulary	76.92 (26)
Word Study	0.00 (5)	World History	100.00 (3)
Writing Strategies	93.67 (332)	Overall	91.34 (1432)

